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ADVANCED ATTACK MITIGATION IN IOT GATEWAY PROTOCOLS

K. Praveen Kumar [a,c,*], Dr. N. Suresh Kumar [b]

[a] Assistant Professor, Department of Information Technology, Kakatiya Institute of Technology and Science, Warangal (KITSW), Telangana, India
[b] Associate Professor, Department of Computer Science Engineering, GITAM Institute of Technology, GITAM (Deemed to be University), Visakhapatnam, India
[c] Research Scholar, Department of Computer Science and Engineering, GITAM University, Visakhapatnam, Andhra Pradesh, India

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ABSTRACT

With the increasing number of users on the internet, numerous cyberattacks are becoming more and more common. Proper detection of these attacks by Intrusion Detection Systems (IDS) is extremely important, particularly for IoT networks. Deep learning methods have proved to be very promising for enhancing IDS performance. This paper presents an end-to-end system for attack detection and prevention in IoT networks with the use of data augmentation, preprocessing, feature extraction, and deep machine learning algorithms. The class imbalance is resolved using the Enhanced Synthetic Minority Over-Sampling Technique (ESMOTE), and preprocessing operations normalize and clean the data for improved model performance. Feature extraction involves statistical features and Shannon entropy-based features, which are fused and sent through a feature selection process. A new 2D-LICM hyper-chaotic map combined with Walrus Optimization (2D-LICMHy-CM_WO) is used to enhance feature selection through enhanced search diversity, convergence rate, and eliminating redundancy. The Dense Convolutional Spatial Attention-based Enhanced Bi-GRU (DCSAtten_EBi-GRU) effectively extracts attack pattern dependencies for precise detection, and an Enhanced Double Deep Q-Learning Network (DoubleDQN) offers dynamic adaptive real-time countermeasures. Experimental findings prove that the proposed solution can obtain a 99.6% detection accuracy with an F1-score of 0.98 and outperforms current IDS models in false positive rate and detection time.

1. Introduction

Internet use is on an exponential rise, and dependence on cloud services is increasing at the same rate. Operations are focused on core businesses. Core business and shifting IT services to the cloud are under the same operations. Several inducements force firms to opt for internet-based services. Malicious traffic has increased sharply (Brindha Devi et al., 2024). Cyberattacks are diverse and wide-ranging. The type of attack includes unauthorized usage of remote data or disrupting services for customers (Nascita et al., 2022; Mihoub et al., 2022). These attacks have great potential to influence a country's economy, economics, national security, and culture. The assaults must be protected from internal and external sources like the government and private organizations (Yoon, 2020; Jhanjhi et al., 2020; Chaganti et al., 2023). Automated techniques are extremely crucial for the rapid and accurate identification of threats.

IDS is a good approach to strengthening system security (Cakir et al., 2020; Khedr et al., 2023). IDS is a security program that tracks communication and notifies of suspicious activity. There are two categories of IDS: Host-based IDS (HIDS) and Network-based IDS (NIDS) (Aslam et al., 2022). HIDS, also referred to as a passive component, focuses on one host (Khedr et al., 2023). NIDS, a mode of an active component, processes network packets and scans for network threats (Aslam et al., 2022). The paper explored the NIDS, which analyzes network traffic to identify anomalous behavior and protect nodes from complex attacks. Effective security response systems prevent and react to invasion attempts from the outside (Popoola et al., 2020). While they vary from signature-based detection to anomaly detection and procedures associated with incident response, many produce different results, and it is tough to choose the correct response to a situation. Since wrong reactions under certain conditions can cause disastrous effects, a quick decision-making process must be followed. It is highly important to evaluate the effects of security response mechanisms in security networks, as there are several underlying reasons for this.

An effect assessment is the only way to tailor response methodologies to particular security risks. Different intrusion scenarios may have to be responded to differently. The customization of security measures is an important requirement for keeping pace with new cyber threats and

* Corresponding author.
E-mail addresses: kpk.it@kitsw.ac.in, pkatukoj@gitam.in (K.P. Kumar), snandiga@gitam.edu (Dr.N.S. Kumar).

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efficiency. Generally, one-size-fits-all approaches rarely respond appropriately to modern cybersecurity nuances. Resource allocation is also a critical aspect. Available cybersecurity resources include the staff, tools, and finances. They are limited and very essential. The impact of the assessment will determine the better use of the resource.

Machine learning (ML) mechanisms within network security have become quite common in enhancing the ability of an IDS to detect attacks through its learning capabilities. They are good defense mechanisms (Wani & Khaliq, 2021). AML is one of the significant challenges for cyber defense in this category (Al-Amiedy et al., 2022). AML has become an effective threat to the success of such systems. An adversarial attacker can degrade network security by using adversarial properties in machine learning methods. Attackers can exploit the vulnerabilities of NIDS by degrading the network traffic (Bhayo et al., 2023). Creating samples is a hostile behavior that degrades the ML algorithm by achieving favorable results for the attacker. These vulnerabilities are crucial to cybersecurity as an undetected breach may compromise all or part of a business (Gaber et al., 2022; Srinivas & Manivannan, 2020). The development of ML-based methods is vital in defense against adversary attacks.

Some ML classifiers do not always reach maximum performance for the multiclass case with more features. Self-learning intrusion detection systems may identify and classify known and unknown threats, hence allowing proactive measures against hostile network activity (Shurman et al., 2020; Susilo & Sari, 2020). DL is a composite subgroup of ML techniques that addresses the constraints of narrow nets. DL methods have been very useful for speech recognition, image processing, natural language processing, and many other related fields. The goal of this paper is to enhance the Network Intrusion Detection System (NIDS) by incorporating advanced attack mitigation strategies (Reddy & Shyam, 2022; Nayak et al., 2021; Duraisamy et al., 2021). Some of the existing optimization techniques, such as Greylag goose optimization (El-Kenawy et al., 2024), Puma Optimizer (Abdollahzadeh et al., 2024), and Football optimization algorithm (El-Kenawy et al., 2024), are employed for the feature selection framework.

Current intrusion detection systems for IoT networks struggle with real-time, multi-class attack detection due to limitations in temporal-spatial feature learning and adaptability to dynamic threats. Existing deep learning-based solutions often suffer from high false positive rates and lack interpretability, making them impractical for resource-constrained IoT environments. This paper addresses these challenges by proposing a novel DCSAtten_EBi-GRU architecture combined with Double DQN-based mitigation, offering enhanced detection accuracy, efficiency, and model explainability. The major contribution of this proposed study is,

- This research created an enhanced SMOTE algorithm for handling class imbalance, in addition to thorough data preprocessing with missing value removal, duplicate elimination, inconsistency resolution, and adaptive min-max normalization to improve model accuracy.
- This model integrated statistical features (mean, median, variance, standard deviation) with Shannon entropy-based features to efficiently cover both distributional characteristics and randomness for better discrimination between normal and malicious traffic.
- Also, it integrated a 2D-LICM hyper-chaotic map with Walrus Optimization to optimize feature selection by enhancing search diversity, increasing convergence, and reducing redundancy.
- Lastly, introduced a two-stage defense mechanism employing DCSAtten_EBi-GRU for precise multi-attack detection and an Enhanced DoubleDQN for real-time, adaptive attack prevention through dynamic traffic management in IoT gateway settings.

The remaining paper includes Section 2 contains related works, Section 3 includes the proposed methodology, Section 4 has results and discussion and Section 5 contains the conclusion and future work.

2. Related Works

Ashiku & Dagli, 2021 seek to design a NIDS that incorporates deep learning techniques. Its goal is to improve the accuracy of detection in different kinds of network attacks using powerful machine learning algorithms. The study sought to enhance the real-time response capabilities of NIDS by investigating the performance of several deep learning models without giving rise to false positives. They also want to show that DL has the potential to automatically detect even more complex infiltration patterns and therefore contribute to the safety framework of networked systems.

Qazi et al., 2023 aimed to fuse various deep learning architectures to tap their complementary capabilities and enhance the performance of detection. Authors could hope that a fusion of multiple feature extraction techniques and classifiers might overcome the constraints of standard NIDS and break the very low detection rate barrier. The paper thus aimed at demonstrating the sufficiency of the hybrid DL-based NIDS HDLNIDS in dependably detecting various types of intrusions into networks by testing its effectiveness on common datasets, consequently improving overall safety measures while providing a feasible solution for current security issues.

Alrayes et al., 2024 suggested a goal to design an intrusion detection system channel attention-based CNN method to develop a CNN model based on the NSL-KDD dataset. It aims to enhance the accuracy and effectiveness of intrusion detection by applying a novel CNN structure that includes an attention mechanism to filter out false positives and raise detection rates for many kinds of attacks. The second aim of this work was to compare the performance of the model with existing methodologies, to obtain some useful insights on using attention channels in deep learning for cybersecurity.

Kshirsagar et al., 2022 described an intrusion detection system that relied on a feedforward LSTM gate approach to notice and categorize attacks. In this context, the paper aimed at enhancing the accuracy of detecting the applications based on the skills of LSTM to process sequential data, even when conventional datasets are being used. The authors aimed to compare their model with other established methods and hence find its advantages in correctly identifying various attacks. Finally, the study searched for a strong solution to develop real-time intrusion detection with an ever-increasing demand for efficient security measures in numerous applications.

Sohi et al., 2021 introduced recurrent neural networks (RNNs) and aimed at boosting the network intrusion detection systems. Scientists hoped that the research could increase the accuracy of attacks and lessen false alarms using deep learning techniques. A researcher tried to show how RNN-based methods might present some temporal relationships in network traffic, so advanced intrusions may be discovered more effectively. The study ended with the advanced state-of-the-art in NIDS through insight into deep learning enhancement for improved cybersecurity.

Al-kahtani et al., 2023 suggested improving intrusion detection for IoT systems by combining deep learning models Gated Recurrent Units and Long Short-Term Memory. The authors aim to utilize the characteristics of both architectures to identify anomalies within the IoT environment with high accuracy. Specifically, it focuses on the specific problems of IoT networks and their limited resources and a multitude of routes of attack. This work was executed solely to test the applicability of the proposed model in real-time intrusion detection applications, thus improving the security of IoT systems.

Rao et al., 2021 introduced a hybrid intrusion detection system approach based on the combination of sparse autoencoder and deep neural network approaches. It was aimed at enhancing the detection accuracy and efficiency obtained through the utilization of sparse autoencoders through unsupervised learning for feature extraction and classification using the application of the deep neural network. The authors hope to show how this hybrid approach could feasibly detect the range of assaults with limited false positives. This study contributed to

Table 1
Existing and proposed approach comparison.

References	Techniques	Advantages	Limitations
Ashiku & Dagli, 2021	Deep Learning	High accuracy, adaptability to new attacks	Requires large datasets, computational overhead
Qazi et al., 2023	Hybrid Deep Learning (HDLNIDS)	Combining the strengths of different models, improved performance	Complexity in model design and training
Alrayes et al., 2024	CNN with Channel Attention	Efficient feature extraction, improved accuracy	May be sensitive to noise in data
Kshirsagar et al., 2022	Feed-forward LSTM	Effective for sequential data, handles long-term dependencies	Computational cost, potential for overfitting
Sohi et al., 2021	RNNIDS	Handles sequential data, captures temporal dependencies	Complexity in model design, computational overhead
Al-kahtani et al., 2023	GRU-LSTM Fusion	Combining the strengths of GRU and LSTM, improved performance	Complexity in model design and training
Rao et al., 2021	Sparse Autoencoder + Deep Neural Network	Feature extraction, dimensionality reduction	Requires careful tuning of hyperparameters
AlJamal et al., 2024	Random Forest classifier for XSS detection in IoT over 5G	High classification accuracy (99.89%), efficient detection of malicious XSS patterns	Validated only on specific XSS cases, limited applicability to other IoT attack types
Al-Na'amneh et al., 2024	CNN-SVM hybrid model for DoS/ DDoS detection	High accuracy and F1-score (99%), superior to individual ML models	Restricted to binary classification (DDoS vs. non-DDoS), no testing on multi-class or unknown attacks
Alsarhan et al., 2024	ABC algorithm for hyperparameter tuning (Decision Trees, SVM, KNN)	Achieved up to 100% accuracy, precision, recall, and F1-scores; improved model generalization	Potential overfitting requires validation on diverse real-world IoT data
Alqura'n et al., 2024	ANN-based models with filter and wrapper feature selection	Very high accuracy (99.84% and 99.79%) across two IoT datasets; statistically validated improvements	Sensitive to initial ANN conditions, performance depends on well-selected features

powerful development in the creation of intrusion detection systems in cybersecurity through assessments of how effective the system is on standard databases.

AlJamal et al., 2024 introduced a Random Forest classifier to identify and forecast XSS attacks on IoT systems running over 5G networks. The technique was to train the model using IoT security data, with an emphasis on detecting malicious XSS patterns. The model registered a high classification accuracy of 9.89%, indicating its efficiency in protecting IoT environments. Yet, the method was only validated on particular XSS attack cases, and thus its applicability to other types of IoT attacks remains limited.

Al-Na'amneh et al., 2024 presented the CNN-SVM hybrid model, which was developed to make the detection of DoS and DDoS attacks more robust in IoT networks. The procedure consisted of applying a Convolutional Neural Network (CNN) to extract the features and a Support Vector Machine (SVM) to classify the extracted features. The model attained a remarkable accuracy and F1-score of 99% on the CIC IoT dataset, which is higher than the performance of individual ML models such as CNN, SVM, KNN, Naïve Bayes, and Logistic Regression. Nonetheless, the model was restricted to binary classification (DDoS vs. non-DDoS) and had no evaluation on multi-class or unknown attack classes.

Alsarhan et al., 2024 discussed that the Artificial Bee Colony (ABC) algorithm was used for hyperparameter tuning in Decision Trees, SVM, and KNN classifiers for IoT malware detection. The procedure was to tune the hyperparameters of these machine learning models to enhance their performance on IoT network traffic data. Consequently, the tuned classifiers attained up to 100% accuracy, precision, recall, and F1-scores in training and testing phases. Nonetheless, the research was constrained by the possibility of overfitting and the necessity for validation on more varied, real-world IoT traffic.

Alqura'n et al., 2024 suggested that the Artificial Neural Network (ANN)-based models were created to identify and block Cross-Site Scripting (XSS) attacks in IoT devices on 5G networks. The procedure included employing filter and wrapper-based feature selection techniques for better model performance and then validating through the NF-ToN-IoT-v2 and Edge-IIoTset datasets. The Bilayered Neural Network (2×10) had an accuracy of 99.84%, while the Trilayered Neural Network (3×10) had a success rate of 99.79%. Limitations were the sensitivity of ANN models to initial conditions and their dependence on well-chosen features. ANOVA tests were employed to verify the consistency and statistical significance of the improvements across feature selection methods. Table 1 indicates the comparison of existing as well as proposed approaches.

2.1. Problem Statement

As IoT networks and devices have expanded at a very high rate, it has become increasingly difficult to ensure correct, real-time intrusion detection under varying, complex, and dynamic cyberattacks. Traditional NIDSs are unable to detect and respond to sophisticated cyberattacks. Hence, this problem exists with every connection. It is amplified by both the volume of network traffic and strategic minds of attackers, through huge false positives and relatively low detection accuracy. Deep learning has emerged as a potential solution for this due to its capability to extract complex features from large volumes of data. Existing machine learning models and traditional ones tend to perform poorly with high false positives, low scalability, and high resource requirements in IoT settings. Moreover, most systems are highly specialized in specific types of attacks or datasets, making them less generalizable and less effective. There exists a strong necessity for sophisticated, explainable, and effective hybrid deep learning models that can precisely identify various attack types in evolving IoT environments. This work responds to these issues by combining spatial attention techniques, improved Bi-GRU structures, and mitigation approaches based on reinforcement learning. In this paper, the authors proposed novel deep-learning techniques that can be applied to improve feature representation and detect known and unknown incursions by combining sparse autoencoders with deep neural networks. The comparative study will be done with the existing approaches to analyze how this methodology performs in accuracy, efficiency, and its potential in dealing with newer threats in the evolving nature of networks.

3. Proposed Methodology

The attack scenario here is emulating a range of network security attacks in an IoT environment. The attacks encompass Distributed Denial of Service (DDoS) attacks, where malicious actors flood a network with heavy traffic to disable service; SQL Injection attacks, which are designed to take advantage of web application vulnerabilities; and malware-based attacks against IoT devices. The attacks are crafted such that they emulate real-world situations when IoT devices are under

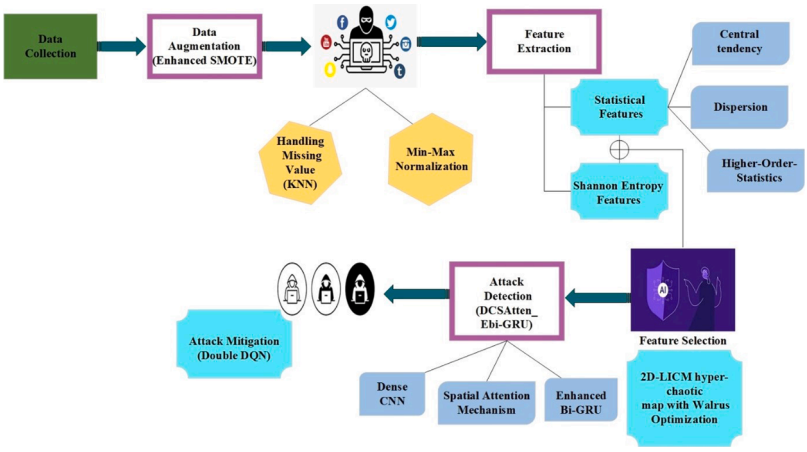


Fig. 2. Architecture of the proposed method.

attack from known threats as well as zero-day attacks. The suggested models are validated for detecting and classifying such attacks accurately with minimal false positives, lending strong support against adaptive cyber threats in IoT networks. The proposed work develops Enhanced SMOTE as a fundamental data augmentation technique for network intrusion detection. Moreover, data preprocessing techniques include aggressive cleaning that eliminates all missing values, duplicates, and inconsistencies to ensure that the data is accurate. Also apply adaptive min-max normalization of numerical parameters, for instance, packet size and duration, to scale these into a common range between 0 and 1. In feature extraction, use statistical features such as central tendency, including means and medians, and central dispersion, including variance and standard deviation, to enhance the differentiation between normal and attack traffic. The randomness and predictability of network packet sizes and timings will be evaluated for anomaly detection employing Shannon entropy-based features.

Then utilize a 2D-LICM hyper-chaotic map in combination with Walrus Optimization to enhance the search diversity and convergence speed in the process of feature selection. For the attack detection task, we employ the Dense Convolutional Spatial Attention-based Enhanced Bi-GRU architecture that integrates the dense convolutional layers with spatial attention processes. Finally, we present an Enhanced Double Deep Q-Learning Network (Enhanced DoubleDQN) framework to mitigate attacks through dynamic blocking or limiting of harmful traffic based on real-time network monitoring. The strategy for adapting the gateway's response to changed attack methods is optimized in this model. Fig. 2 signifies the architecture of the proposed approach.

3.1. Data collection

The CIC IoT Dataset 2023 is an extensive dataset focused on network traffic generated from the Internet of Things (IoT), aimed to be utilized to develop machine learning models in cyber security applications. It captures various attack scenarios, including DDoS, brute-force, infiltration, and data exfiltration besides malicious and benign network traffic to simulate real-world IoT environments. The IoT devices might be smart thermostats, security cameras, or even smart plugs-thus, known for the generation of both normal and harmful traffic patterns.

The dataset comprises over several million entries and contains source/destination IPs, port numbers, packet counts, timestamps, and derived statistical measures. The CIC IoT Dataset has some challenges, like skewed class distribution, with attack types like SQL Injection and Malware less represented than DDoS. It also has noise and inconsistencies in network traffic data, such as anomalous packet sizes and times. There are also missing and incomplete values in critical fields that can impact feature quality. These problems adversely impact model training and detection performance if not addressed appropriately. To address this, the solution strategy utilizes aggressive data cleaning, adaptive normalization, and Enhanced SMOTE for balanced, trustworthy learning. CIC IoT Dataset 2023 will, therefore, allow the researchers to train, test, and evaluate the models for anomaly detection, attack mitigation, and analysis, ensuring threat prevention as well as detection in complex IoT networks.

3.2. Data Augmentation Using Enhanced SMOTE

Enhanced SMOTE is more of an advanced version of the original SMOTE technique where synthetic minority class samples are used for over-sampling their respective classes to resolve the problematic of class inequity in datasets. Its objective is to perform and optimize data distribution while reducing model bias during training. The improved SMOTE eliminates overfitting, variety diminution, and lesser sample generation around the decision limits.

In Enhanced SMOTE, minority and majority class samples are weighted according to the density. The generation formula is denoted by,

$$y_{new} = y_j + \alpha \times \lambda \times \left(y_{mm} - y_j\right) \tag{1}$$

Where α is an adaptive weight that changes based on the sample densities around original minority class sample selected for oversamplimng (y_j), y_{new} signifies the newly generated synthetic data point, y_{mm} is the nearest neighbor of y_j within the minority class, and λ denotes a random value in the range [0,1]. It assures that denser regions produce fewer new points, whereas sparse regions are emphasized. Due to the high disparity between benign and minority attack classes, the Enhanced Synthetic Minority Over-Sampling Technique (ESMOTE) was applied. ESMOTE generates synthetic minority samples based on adaptive density weights to balance the dataset, reduce bias, and enhance the model's detection accuracy.

3.3. Pre-processing

Data preprocessing is an important pre-processing stage in preparing raw data for analysis, as it ensures data is clean, consistent, and available for DL models or another kind of analysis. Accurate preprocessing could thus improve the quality of results by removing noise, filling gaps, and standardizing data formats.

3.3.1. Handling Missing Values

Deletion of rows containing missing values; mean, missing data, median, or mode as imputation of missing values; more elaborate imputation techniques, such as K-Nearest Neighbors (KNN) or regression-based imputation. The missing values in the dataset were

imputed using **mean substitution**, replacing missing entries with the average value of their respective feature. Here, y_{new} represented as the imputed value replacing the missing data point, y_j indicates the non-missing values in the feature, m is the total number of non-missing data points in the feature.

$$y_{new} = \frac{\sum_{j=1}^{m} y_j}{m} \tag{2}$$

Duplicate records were detected using a row-wise duplicate check and subsequently removed to ensure the dataset's uniqueness and consistency.

3.3.2. M-M Normalization

To bring all features into a consistent scale, Min-Max normalization was applied, and it scales numerical features toward a pre-specified variety generally among 0 to 1, but the transformation depends on several data characteristics, such as variable distributions and feature importance. It is calculated using the formula: where,

$$y_{norm} = \frac{y - y'_{min}}{y'^{min}_{max}} \tag{3}$$

Where, y represent the original value of the feature, y_{norm} is the normalized value, $y'_{min}$and y'_{max}indicates the shifted minimum and maximum which could be more than the actual minimum and maximum that might outline the dataset excluding outliers or because of constraints of features.

3.4. Feature Extraction

Data feature extraction is an important step in machine learning and data mining. It is about transforming raw data into descriptive features that algorithms can use to identify patterns and make predictions. Statistical features such as mean, variance, and standard deviation were extracted to capture the distributional properties of network traffic flows.

3.4.1. Central Tendency

Ø Mean: The average value of a property, aids in identifying typical behavior.

$$Mean = \frac{1}{M}\sum_{j=1}^{M} y_j \tag{4}$$

Where, $y_j$are the data points and Mindicates the total number of observations.

Ø The median is the center value in a dataset, which is less impacted by outliers than the mean.

3.4.2. Dispersion

Ø Variance: This represents the spread of data about the mean, indicating how much values deviate.

$$\text{var} = \frac{1}{M}\sum_{j=1}^{M}\left(y_j - mean\right)^2 \tag{5}$$

Ø Standard Deviation: The square root of variance, which represents the degree of departure from the mean.

3.4.3. Higher-Order Statistics

Ø Skewness: It is the degree of asymmetry to one side in a distribution of data as opposed to the other.

$$Skew = \frac{1}{M}\sum_{j=1}^{M}\left(\frac{y_j - mean}{SD}\right)^3 \tag{6}$$

Ø Kurtosis: This pertains to how "tailed" a distribution is. High kurtosis means there is a lot of data in the tails, potentially indicating the existence of outliers. Here, SD tends to standard deviation.

$$Kurt = \frac{1}{M}\sum_{j=1}^{M}\left(\frac{y_j - mean}{SD}\right)^4 - 3 \tag{7}$$

A kurtosis number greater than zero implies a leptokurtic distribution, whereas a value less than zero indicates a platykurtic distribution.

3.4.4. Feature Extraction in Shannon Entropy (ShEn)

Shannon entropy-based features were computed to quantify the randomness and unpredictability in traffic data, which are often indicative of abnormal or malicious activity. The combination of these features enriches the feature space by incorporating both quantitative and information-theoretic attributes, improving detection capability. ShEn represents a set of related variables that change linearly depending on the logarithm of various possibilities. This metric is useful for determining the dynamic order of a system. ShEn depends on the addition principle of the composite net, which states that if a system is partitioned into two independent components ($B$and A), the additivity law applies.

$$S(B,A) = S(B) + S(A) \tag{8}$$

The entire entropy of the net is represented by the equation $S(B,A)$, and the entropy of subsystems $B$and $A$are represented by $S(B)$and $S(A)$. ShEn positively notices complex nets with short-range interactions. The physical division of a network into subnetworks demands that the subnets be isolated in space to avoid residual contact and correlation. A long-range interaction-dominated system cannot be made statistically independent through spatial departure since the effect of the communication is preserved for any distance leaving it correlated at all distances. ShEn performs very poorly on non-extensive (non-additive) systems with long-range interactions. When there exist large numbers of domains, the corresponding entropy level is overstated. Moreover, the temporal relationship of the time series signal for the retrieved values is still unknown.

3.4.4.1. Concatenation. The statistical and Shannon-based entropy properties are extracted to capture significant network traffic characteristics, such as central tendency, variability, and randomness. Such attributes allow one to gather vital information in terms of normal and attack patterns. After extraction, they are fused together to create a holistic feature set that, after undergoing feature selection, helps improve the model's ability to concentrate on the greatest important quality.

3.5. Feature Selection Using 2D-LICMHy-CM_WO

A 2D-LICMHy-CM_WO for feature selection boosts the diversity of search and reduces the convergence time. The randomness involved in introducing this map prevents features from converging too fast; still, Walrus Optimization harmoniously balances exploration by exploitation to improve the accuracy in the choice of features and increase the computational time efficiency. At the feature selection phase, the 2D-LICM hyper-chaotic map with Walrus Optimization offers feature

Algorithm 1
2D-LICMHy-CM_WO for Feature Selection.

Input:
Objective function $f(Y)$
Number of walruses M
Number of decision variables n
Maximum iterations r_{max}
Variable bounds $[ka_i,\ oa_i]$
2D-LICM parameters $(\mu,\ \alpha,\ \beta,\ \gamma)$
Initialize:
Generate initial chaotic sequences $x_0,\ y_0$ using 2D-LICM in Eq. (9)
Randomly initialize positions Y_j of M walruses within variable bounds
Evaluate initial objective values $E_j = f(Y_j)$
Identify the strongest walrus PV with highest E_j (best solution)
For r = 1 to r_{max} do:
Update 2D-LICM sequences (x_n, y_n)
For each walrus j = 1 to M do:
Exploration Phase
Generate new position using Eq. (12)
Compute $E_j^{(q1)}$
If $E_j^{(q1)} < E_j$
Update $Y_j \leftarrow Y_j^{(q1)}$
Exploitation Phase
Generate new position by Eq. (13)
Compute $E_j^{(q3)}$
If $E_j^{(q3)} < E_j$
Update $Y_j \leftarrow Y_j^{(q3)}$
Update the strongest walrus PV with the best current E_j
End For
Output:
Best-found solution PV and its corresponding objective value

importance ranking by measuring the contribution of each feature to classification performance and convergence behavior. The ranking allows practitioners to know which network features (e.g., packet size, entropy, timing patterns) have the most significant impact in discriminating between normal and attack traffic.

3.5.1. 2D-LICM hyper-chaotic map

Thus, the exploration-exploitation balance can be improved by efficiently merging the Walrus Optimization approach with the 2D-LICM, a hyper-chaotic map. This dynamic behavior allows retention of diversity, and the algorithm will not get stuck in local optima.

The 2D-LICM produces chaotic sequences based on the logistic and Chebyshev maps controlled by an inertia factor. The formulas of the 2D-LICM are as follows:

$$\begin{aligned} y_{n+1} &= \mu.y_n.(1-y_n) + \alpha.\sin(x_n) \\ x_{n+1} &= \cos(\beta.x_n) + \gamma.(1-\cos(y_n)) \end{aligned} \tag{9}$$

Variables for the n^{th}iteration: y_n and x_n, Its parametersμ, α, β, and $\gamma$are relevant to chaotic behavior, the logistic term $\mu.y_n.(1-y_n)$ is responsible for the chaos, whereas the inertia factors ($\alpha and \gamma$) along with sinusoidal elements retain the hyper-chaotic dynamics.

3.5.2. Walrus Optimization (WOA)

Walruses are big marine animals that live natively in the Arctic. Their appearance can be easily identified with the big tusks and wrinkled skin. Despite their fearsome size, walruses are friendly creatures. These social animals are generally found in huge groups together. Arctic ecologies find a very important role in such beautiful creatures who are known for their intelligence and flexibility.

3.5.2.1. Initialization Phase. WOA is a metaheuristic search algorithm using walruses as searchers. In WOA, every walrus would be the solution to the problem of optimization. Every walrus position determines possible values to variables in the problem in the search space. All walruses represent a vector, as well as the population should be exactly represented by a population matrix. Population with random initialization of walruses is undertaken upon the implementation of WaOA.

$$Y = \begin{bmatrix} Y_1 \\ \vdots \\ Y_j \\ \vdots \\ Y_M \end{bmatrix}_{M\times n} = \begin{bmatrix} y_{1,1} \cdots y_{1,j} \cdots y_{1,n} \\ \vdots \ddots \vdots \ddots \vdots \\ y_{j,1} \cdots y_{j,i} \cdots y_{j,n} \\ \vdots \ddots \vdots \ddots \vdots \\ y_{M,1} \cdots y_{M,i} \cdots y_{M,n} \end{bmatrix}_{M\times n} \tag{10}$$

The equation follows: Yrepresent a population of walruses, $Y_j$represents a candidate solution among the list of walruses, $y_{j,i}$represents the cost of the i^{th}choice adjustable proposed through the jth walrus, M indicates the amount of walruses, and $n$represent the number of DV.

Every walrus gives a choice in solving the issue as well as its recommended values for the DV can be set up to test the objective function. The below equation gives approximate ranges for the detached variable results from seahorses.

$$E = \begin{bmatrix} E_1 \\ \vdots \\ E_j \\ \vdots \\ E_M \end{bmatrix}_{M\times\ 1} = \begin{bmatrix} E(Y_1) \\ \vdots \\ E(Y_j) \\ \vdots \\ E(Y_M) \end{bmatrix}_{M\times\ 1} \tag{11}$$

The objective purpose vector Eas well as its value $E_j$is tested for the j^{th}walrus. OF values give the finest idea of the superiority of candidate keys. The best member is that candidate solution which maximizes the value for the objective function. The worst member is the answer which gives the smallest value for the goal function. Goal function values are perturbed at each cycle so that the best and worst members get affected.

3.5.3. Exploration Phase

It feeds on a diverse range of benthic marine animals like tunicates, sea cucumbers, shrimp, soft corals, mollusks, and tube worms. Walruses are known to consume benthic bivalve mollusks such as clams, but the favorite ones are benthic bivalves. The walrus uses its flippers and sensitive whiskers to feel its way around and graze on the seabed. The solidest walrus through the longest tusks leads the pack which the rest of the members wait for. It is then well known that the length of walrus tusks correlates with the quality of candidate solutions in terms of objective function values.

The strongest among them is the one with the largest price for the OF. The search activity by walruses produces new sections of the search arena, thereby enhancing the power of exploration of the global WaOA search. The feeding mechanism of walruses is applied to replicate the location by following the most important member of the group.

$$\begin{aligned} y_{j,i}^{q_1} &= y_{j,i} + rand_{j,i}\cdot\left(PV_i - J_{j,i}\cdot y_{j,i}\right), \\ Y_j &= \begin{cases} Y_j^{q1}, E_j^{q1} < E_j \\ Y_j, else \end{cases} \end{aligned} \tag{12}$$

Where Y_j^{q1}represent the new produced location for the jth walrus, obtained according to the first stage, its i^{th}dimension being $y_{j,i}^{q_1}$, it's OF charge being E_j^{q1}, $rand_{j,i}$being casual numbers in the range [0, 1], and PV indicates the strongest walrus with the greatest cost of the objective.

3.5.4. Exploitation Phase

Polar bears and killer whales attack walruses rather often. The strategy of evading and counterattacking the attackers changes the location of walruses in the neighborhood. Simulation of natural behavior also enhances the use of local problem space by WaOA to find a set of candidate solutions.

$$y_{j,i}^{q_3} = y_{j,i} + \left(ka_{local,i}^r + \left(oa_{local,i}^r - rand\cdot ka_{local,i}^r\right)\right) \tag{13}$$

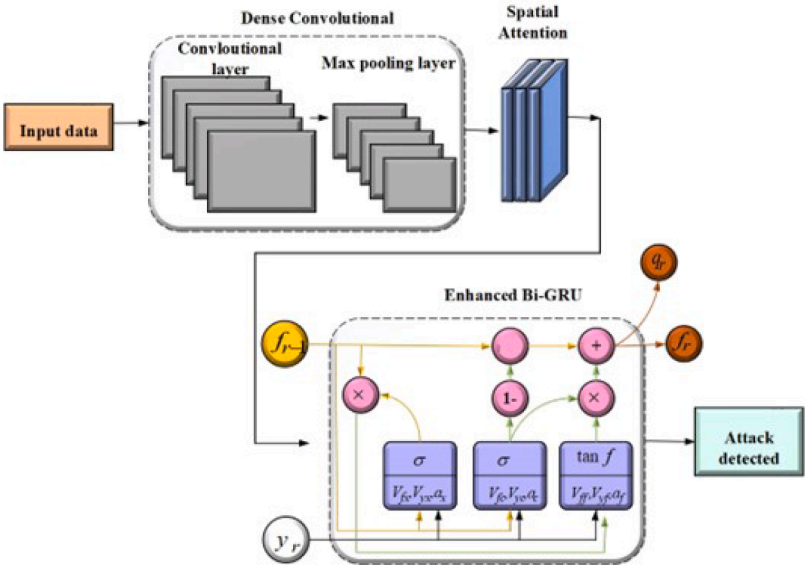


Fig. 3. The Proposed Attack detection model.

$$Y_j = \begin{cases} Y_j^{q_3}, E_j^{q_3} < E_j \\ Y_j, else \end{cases} \tag{14}$$

In this equation, $Y_j^{q_3}$ represents the newly produced position for the j^{th}walrus depend on the 3rd stage, $y_{j,i}^{q_3}$ is its i^{th}measurement, $E_j^{q_3}$ is its OF range, $r$indicates the repetition contour, and $ka_i$and oa_iindicates the lower as well as upper bounds of the i^{th}variable. Algorithm 1 shows the pseudocode for the feature selection using 2D-LICMHy-CM_WO model.

3.6. Attack Detection Using DCSAtten_EBi-GRU

It uses dense convolutional networks with spatial attention techniques to improve feature extraction and temporal learning in the Dense Convolutional Spatial Attention-based Enhanced Bi-GRU architecture. It appropriately captures both spatial linkages and temporal dependencies, making it suitable for tasks involving spatial properties of sequence data, such as time anomaly detection, series forecasting, and natural language processing.

The time complexity of the Spatial Attention and Dense Convolutional modules increases linearly with the feature map size, and the Bi-GRU module with variational dropout adds a time complexity of $O(T \times H^2)$ for each sequence, with T denoting the sequence time steps and H being the number of hidden units. While these layers contribute computational overhead, the architecture is still appropriate for edge computing and mid-tier IoT gateways where decent processing capabilities are present. Fig. 3 shows the architecture of proposed attack detection model.

3.6.1. Dense Convolution Neural Network (DCNN)

The DCNN captures spatial characteristics by connecting one layer to the next in a feed-forward manner, encourages feature reuse, and prevents vanishing gradients.

$$a_k = G_k([x_0, x_1, ..., x_{k-1}]) \tag{15}$$

Where, $a_k$is the output of the k^{th}layer, $G_k$is a transformation, such as convolution, activation at layer k. It is a concatenation of all outputs of preceding layers $x_0, x_1, ..., x_{k-1}$.

3.6.2. Spatial Attention Mechanism (SAM)

The SAM generates spatial attention maps that emphasize the most important spatial areas in the feature space. Visualizing the maps enables practitioners to understand what parts of the input data (features or patterns) the model gives priority to when making detection decisions. It contains the SAM, which is known as the convolutional block attention module for classification and detection. SA creates a spatial attention map based on feature relationships. SA put on max-pooling as well as average-pooling laterally the network axis to create an effective feature signifier for spatial attention calculation. The input feature is $E \in T^{G\times V\times D}$forwarded through channel-wise max-poling and average-pooling to yield outputs $E_{nq}^p \in T^{G\times V\times 1}andE_{bq}^p \in T^{G\times V\times 1}$, respectively. The spatial attention map is generated using a convolutional layer and the Sigmoid activation function on the concatenated feature descriptor$N^2(E) \in T^{G\times V\times 1}$.

$$\begin{aligned} E^p &= E{\cdot}N^p(E) \\ &= E{\cdot}\sigma\left(e^{7\times 7}([\max pool(E); avgpool(E)])\right) \\ &= E{\cdot}\sigma\left(e^{7\times 7}\left(\left[E_{nq}^p; E_{bq}^p\right]\right)\right) \end{aligned} \tag{16}$$

Where, $e^{7\times 7}(\cdot)$indicates a convolution process using a kernel scope of 7, as well as $\sigma(\cdot)$signifies the Sigmoid role. This CSA module aims to capture critical spatial features using some weights that amplify these critical parts of the input. Here, the convolutional filters are used in creating an attention map that captures spatial correlations between the feature maps.

$$N_p = \sigma(Conv_{3\times 3}([avgpool(E), \max pool(E)])) \tag{17}$$

Where, N_p represent the spatial attention map, σ signifies the sigmoid activation function. Then the feature map is enhanced by,

$$E_{enhan} = N_p \odot E \tag{18}$$

Where, E_{enhan}refers to the spatially improved feature map, while $\odot$represents element-wise multiplication. Fig. 4 indicates the spatial attention mechanism.

3.6.3. Enhanced Bi-GRU

A variant of dropout in Bi-GRU that improves model regularization is the application of the same dropout mask over all time steps of the sequence with the guarantee that temporal dependencies are maintained. Variational dropout maintains consistent patterns of dropped neurons through an input sequence. Such a method reduces overfit and at the same time enhances the ability of the model to learn robust temporal features, especially useful for jobs such as sequential data analysis, such as network attack detection.

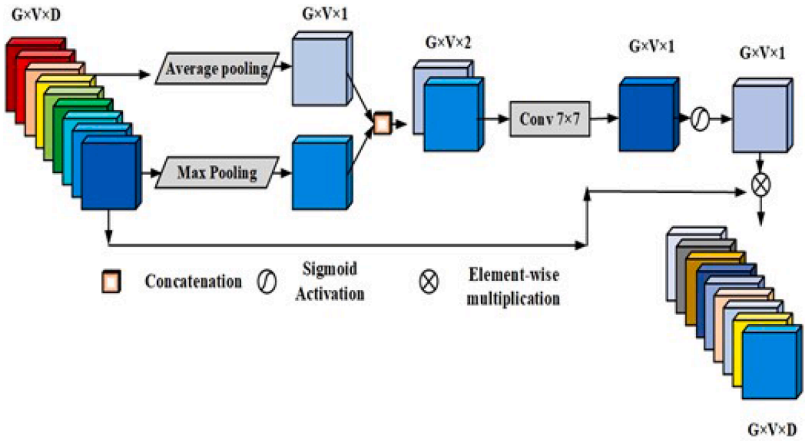


Fig. 4. Spatial attention.

3.6.3.1. Variational Dropout. Variational dropout does better than just one consistent dropout mask for all time steps. Therefore, if a neuron were removed at the start of a sequence, it would drop within the whole sequence so that consistent patterns were learned. The formulation reads as follows:

$$g_r^{'} = g_r.mask, \forall r \in \{1, 2, ...R\} \tag{19}$$

Here,$g_r$signifies the hidden state vector in the time step r, *mask*provides a binary variable sampled once with a Bernoulli distribution Bern and kept unchanged for all Rtime steps. Take the forward pass of a Bi-GRU. Let $y_r$be the inputs at the time step r, and let G_{r-1}be the hidden state from the previous step. A standard forward pass of a GRU includes the following:

$$\begin{aligned} c_r &= \sigma(V_c y_r + O_c g_{r-1} + a_c) \\ t_r &= \sigma(V_t y_r + O_t g_{r-1} + a_t) \end{aligned} \tag{20}$$

Where, $c_r$represents the update gate, t_r provides the reset gate, V_c, O_c, V_t, O_c are weight matrices, and a_c, $a_t$are biased terms.

$$\begin{aligned} \widetilde{g}_r &= \tan g\left(V_g y_r + O_g(t_r.g_{r-1}) + a_g\right) \\ g_r &= (1 - c_r)\cdot g_{r-1} + c_r.\widetilde{g}_r \\ g_r^{'} &= (1 - c_r)\cdot(g_{r-1}.mask) + c_r.\widetilde{g}_r \end{aligned} \tag{21}$$

Where, *mask*constitutes a fixed binary vector that is sampled only at the beginning as well as applied throughout all time steps (r). The symbol ·represents element-wise multiplication. Overall, variational dropout improves Bi-GRU performance, particularly in applications where temporal dependencies must be maintained, such as speech recognition, time series analysis, or network security threat detection.

3.7. Double DQN for mitigation

Double DQN, is an extension of the DQN method: specifically, it addresses common overestimation of Q-values in regular DQNs by splitting up the action selection and assessment processes. It applies well to cases where the correct choice to be made is critical, like cybersecurity mitigation tactics that attempt to dynamically respond to intrusions and threats in a network. The DoubleDQN model has a time complexity of $O(|A| \times |S|)$ per Q-value update, where $|A|$ represents the size of the action space and $|S|$ represents the size of the state space. Training complexity is greater, but the inference process is computationally efficient, supporting real-time mitigation decisions on IoT edge nodes.

In modern DQN, the Q-learning update equation for Q-values, which represent the predicted rewards, is as follows:

$$S(p, b) = S(p, b) + \alpha\left(t + \gamma \underset{b'}{max} S(p', b') - S(p, b)\right) \tag{22}$$

Where, $S(p, b)$represent the Q-value for the move b in state p, α indicates the learning rate, t indicates the immediate reward obtained after taking an action b, γrepresent the discount factor, $max_{b'}S(p', b')$it is the maximum Q-value.

The Q-values of the DoubleDQN model can be examined to see what actions (block, allow, limit) are preferred with varying traffic patterns or threat levels. This gives actionable feedback to cybersecurity operators on the model's dynamic preference of response under real-time attack scenarios. Double DQN architecture can be a very reliable solution in scenarios that require high-precision, flexible, and economical responses for dynamic events caused by cyberattacks, and its ability to learn as well as adapt makes it a highly useful tool in automated systems for security mitigation.

The approach involves an in-depth security analysis to evaluate the robustness of the suggested attack detection models. This includes checking the resilience of the models against adversarial attacks and testing whether the models can sustain detection accuracy when given corrupted or manipulated data. Further, the analysis checks the capacity of the models to detect sophisticated intrusions, including stealth or mimicry attacks. Their performance is also evaluated in stress scenarios such as DDoS attacks and network saturation to check for their stability. Lastly, the integrated mitigation are evaluated in order to test not only detection accuracy but also whether the models are flexible enough to adjust and respond swiftly to attacks in real-time IoT settings.

4. Result and Discussion

4.1. Experimental Setup

The proposed work is compared with other existing methods like LSTM, GRU, Att_CNN, and RNN. Table 2 shows the system configuration.

4.2. Performance calculation and Comparison Analysis

Any system, process, or individual will measure performance through efficiency in terms of speed, accuracy, and effectiveness. Through comparison analysis, the relative strengths, shortcomings, and areas for improvement of several systems, processes, or persons are identified.

4.2.1. Training and validation accuracy

Fig. 5 plots a training and validation curve over ten epochs for a ML approach. The blue line designates the model's precision on the training data, as well as the orange arrow is the model's accuracy on a separate data set known as the validation set. Both training and validation increase at every epoch. However, after a certain point in the training

Table 2
System Configuration.

Device name	DESKTOP-7JRK4SH
Processor	Intel(R) Core (TM) i3-4170 CPU @ 3.70GHz 3.70 GHz
Installed RAM	8.00 GB
Device ID	E4E81576-2C07-414D-8820-64E1B103D0D0
Product ID	00331-10000-00001-AA893
System type	64-bit operating system, x64-based processor
Pen and touch	No pen or touch input is available for this display



Fig. 5. Training and Validation accuracy.

accuracy rises, and the validation accuracy starts to plateau or even decline considerably. This therefore implies that the model begins overfitting the training data, suggesting that it learns the training data too well and does not generalize well in cases of other new data.

4.2.2. Training and validation loss

From Fig. 6, note that training and validation losses decrease as epochs increase. This indicates that the model learns from the data and improves its performance. However, at some epochs, one can see the peaking of validation loss, at which either it stops falling or even increases, while the training loss just keeps falling. This is a warning sign that the model overfits the training information.

4.2.3. Comparison analysis of Accuracy, recall, f1 score, precision, specificity

As it was stated before, the methods described above are broadly applied to many sequential and time-series data processing jobs. However, some drawbacks in the assessment of some of the identified parameters such as accuracy, precision, recall, F1 score, and specificity of this model have been identified. Although LSTMs can take care of long-term dependencies, they often do not pay much heed to a slight temporal pattern in big datasets that causes low precision and accuracy. Since GRUs are also computationally more efficient than LSTMs, there may be some subtle patterns in their data that could cause problems with recall and F1 scores. Traditional RNNs, missing the gated features in LSTMs and GRUs, are much more prone to failures such as vanishing gradients, and thus, perform sub-optimally in all the evaluation parameters.

Regarding the alternatives considered, attention-based CNNs use attention methods that enhance feature extraction but are, however computationally expensive and thus inappropriate for use in real-time applications and can also compromise specificity due to overfitting on noisy data. The approach proposed outperformed others in the achieved metrics, which efficiently resolved difficulties such as disappearing

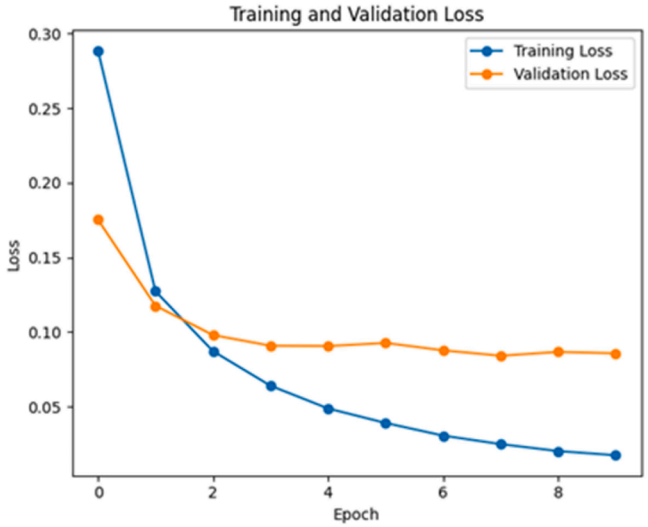


Fig. 6. Training as well as validation loss.

gradients and improving temporal and spatial pattern learning. Fig. 7 shows the comparison analysis of proposed and existing approaches

4.2.3.1. Error Metrics. LSTMs and GRUs are capable of learning temporal dependencies, may miss subtle patterns and suffer a higher FNR, whereas Att_CNNs tend to overfit because of excessive attention on noise, inflating FPR. The traditional RNN suffers from the problem of vanishing gradients; it puts a limit on the capability in processing long sequences, and high values for both FPR and FNR accordingly. Instead, the presented approach significantly reduces both FPR and FNR through improved feature extraction, selection, and adaptive learning methods, which can surely ensure accurate detection. Fig. 8 indicates the FNR and FPR comparison. Table 3 summarizes the comparison analysis.

Table 4 shows comparisons between the training time and test time of different models. The most extended training time is realized for the proposed model, with the RF model realizing the least. Overall, the models have a mixture of performance as far as being efficient computationally, with variations between training and testing time within the different strategies.

Table 5 provides a comprehensive breakdown of model underperformance by various attack types, focusing on accuracy, precision, recall, and F1-Score as important indicators. Certain models like the Proposed Model, LSTM, GRU, and others were found lacking in identifying DDoS, SQL Injection, and Malware-based attacks, typically because they are unable to deal with large volumes of traffic, tiny payloads, or insidious patterns. These weaknesses are due to overfitting, classification errors of legitimate traffic, and inabilities to generalize infrequent patterns of attacks.

4.2.3.2. Processing Time. This implies that the approach or process being examined has linear time complexity, which means that its processing time is proportionate to the magnitude of the input data. Such a relationship is typically seen in algorithms that perform simple operations on every input element, including searching or sorting. When the number of inputs increases, the processing time also increases. A simple linear search algorithm, for example, iterates through every component of an array in search of a target value, with processing time increasing linearly as the number of elements within the array. Similarly, a bubble sort method compares and swaps neighboring members until the array is sorted, with the number of comparisons and swaps increasing linearly as the array's size grows. Fig. 9 shows the processing time of the proposed approach.

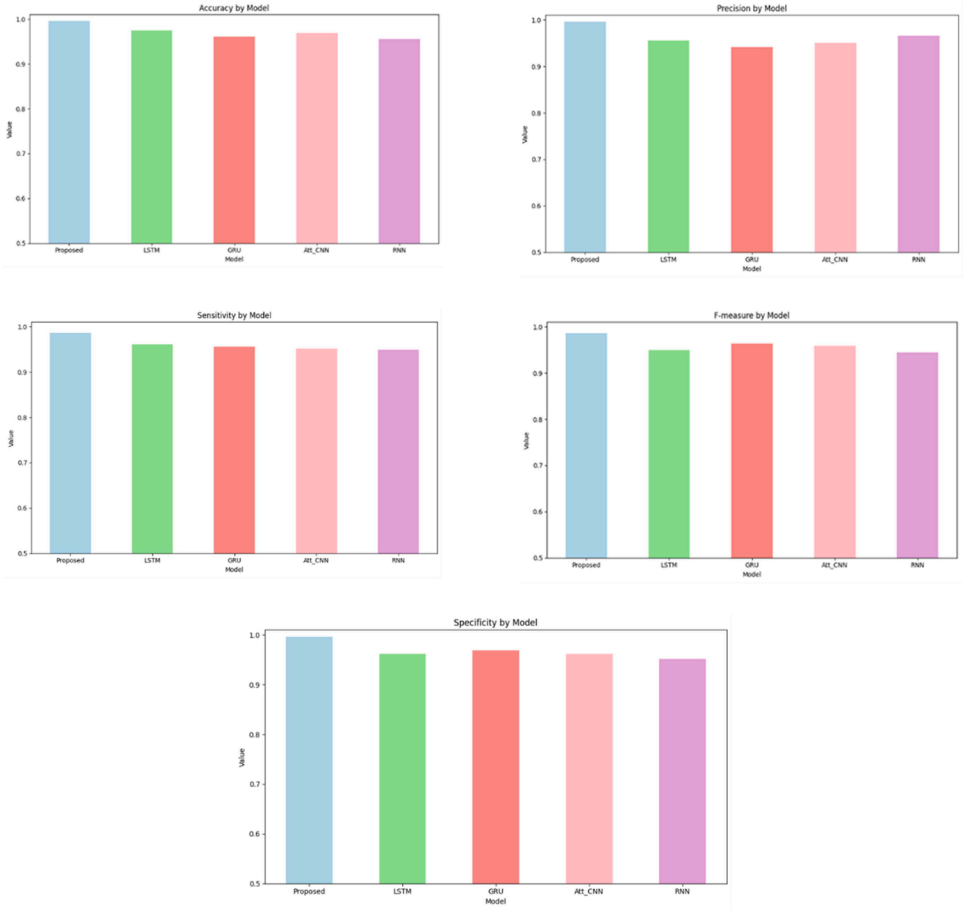


Fig. 7. Proposed and existing techniques comparison.

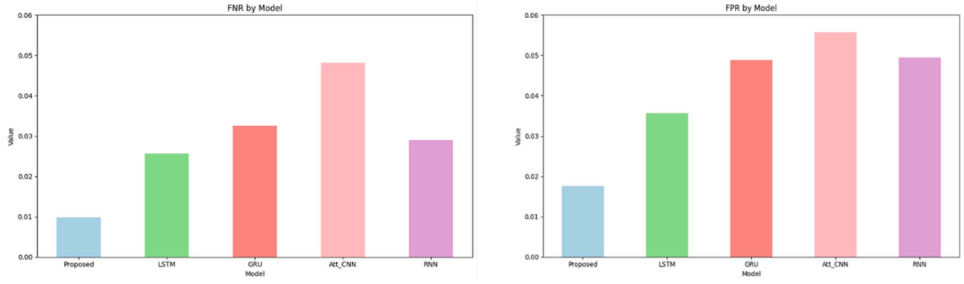


Fig. 8. Error rate comparison.

Table 3
Comparison Analysis.

Model	Accuracy	Precision	Sensitivity	Specificity	F-measure	FPR	FNR
Proposed	0.996	0.996	0.98591	0.99627	0.98564	0.01753	0.00984
LSTM	0.97	0.956	0.96112	0.96183	0.95017	0.03563	0.02565
GRU	0.960	0.941	0.95582	0.96858	0.96337	0.04891	0.03258
Att_CNN	0.968	0.950	0.95183	0.96135	0.95881	0.05571	0.04816
RNN	0.955	0.966	0.94837	0.95159	0.94418	0.04951	0.02896
RF	0.980	0.975	0.97052	0.98247	0.97365	0.02534	0.01984
CNN-SVM	0.973	0.963	0.96782	0.97438	0.96572	0.02974	0.02247
ANN	0.960	0.949	0.95714	0.96242	0.95213	0.04532	0.03527

Table 4
Comparison of Training Time and Testing Time.

Model	Training Time (s)	Testing Time (s)
Proposed	250	15
LSTM	120	10
GRU	110	8
Att_CNN	150	12
RNN	100	9
RF	80	7
CNN-SVM	200	14
ANN	180	13

4.2.4. ROC curve

A receiver operating characteristic curve represents a graphical plot that enables visualization of a binary classification techniques performance. The x-axis displays the FPR, or percentage of negative cases wrongly detected as constructive. The y-axis illustrates the TPR, or quantity of positive events accurately detected.

The ROC curve of the image represents how sensitive is the trade-off of the model to specificity. A point on the curve describes a threshold level for a specific value in the model's categorization decision. As the threshold is lowered, the model gets more sensitive; it has a greater likelihood of correctly categorizing positive cases. However, this increases the number of false positives. Increasing the threshold makes the model more detailed and reduces the chances of negative cases being classified mistakenly as positive; however, this drops the true positive rate. The diagonal line in the image represents the performance of a random classifier, which has TPR equal to FPR everywhere. A model whose ROC curve lies above this diagonal line is better than the random classifier. The area under the ROC curve is a common metric to evaluate how well the given classification model performs. A higher value of AUC indicates better performance. Fig. 10 indicates the ROC curve of the recommended study.

5. Conclusion and Future Study

This proposed study introduced an Enhanced SMOTE, which is effective in balancing samples of the attack and benign datasets for use in the modeling in a bid to reduce the effect of these unbalanced datasets. Data cleaning processes together with Min-Max normalization increase the quality and consistency of the data imported and used in training. The computation of statistical and Shannon entropy-based metrics makes it possible to distinguish between normal and malicious communications. Additionally, the 2D-LICM hyper-chaotic map integrated with Walrus Optimization provides feature selection. However, the Enhanced Bi-GRU model with Dense Convolutional Spatial Attention and a Double Deep Q-Learning Network enhances the accuracy of detection and dynamic traffic management. The proposed system has robust potential for securing IoT but suffers from deployment-related issues such as latency, scalability, and limited resources. Incremental and federated learning approaches will be pursued in future research to accommodate future attacks. A combination of DoubleDQN and optimization methods may increase mitigation efficiency. Lightweight encryption integration will increase communication security. Extended benchmarking and real-time testing are anticipated to ensure real-world performance verification.

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Table 5
Error Analysis of Various Models: Underperformance Metrics, Attack Types, and Reasons for Misclassification.

Model	Attack Type	Underperformance Metric	Value	Reason for Underperformance
Proposed	DDoS	Accuracy	95%	Struggles with high-volume DDoS traffic leading to false negatives.
	SQL Injection	Precision	88%	The model fails to capture subtle patterns of SQL Injection in traffic, causing false positives.
LSTM	SYN Flood	Recall	89%	LSTM fails to recognize traffic bursts in SYN Flood attacks, leading to missed detections.
	Malware-based	F1-Score	83%	The LSTM model struggles with infrequent malware attacks that have small payload sizes.
GRU	DDoS	Specificity	90%	GRU misclassifies benign traffic as DDoS due to similarities in traffic patterns.
	SQL Injection	Precision	84%	GRU struggles with detecting the nuanced patterns of SQL Injection in a mixed traffic stream.
Att_CNN	SYN Flood	Recall	85%	Att_CNN has difficulty detecting SYN Flood attacks due to their bursty nature and low signal.
	Malware-based	Specificity	87%	False positives occur when Att_CNN misinterprets malware as benign traffic due to data fluctuations.
RNN	Buffer Overflow	Accuracy	92%	RNN fails to generalize to unusual or rare buffer overflow patterns within IoT devices.
	SQL Injection	F1-Score	86%	Limited ability to capture the full context of SQL Injection attacks, leading to a lower F1-Score.
RF	SYN Flood	Precision	80%	Random Forest misidentifies benign SYN traffic as a flood attack due to its fixed feature set.
	DDoS	Specificity	78%	Random Forest has issues differentiating normal traffic from DDoS during heavy network congestion.
CNN-SVM	DDoS	F1-Score	81%	CNN-SVM model does not handle high-volume DDoS attacks effectively, leading to underperformance in precision.
	Malware-based	Recall	75%	CNN-SVM struggles with low-frequency malware patterns, leading to missed detections.
ANN	Buffer Overflow	Accuracy	90%	ANN underperforms with rare buffer overflow attacks due to overfitting on more common patterns.

(*continued on next page*)

Table 5 (*continued*)

Model	Attack Type	Underperformance Metric	Value	Reason for Underperformance
	DDoS	Precision	85%	ANN struggles with distinguishing complex DDoS attacks from benign traffic, leading to misclassification.

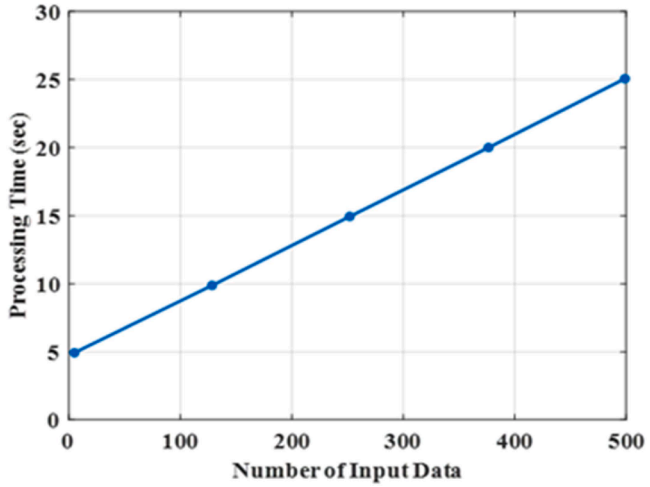


Fig. 9. Processing time.

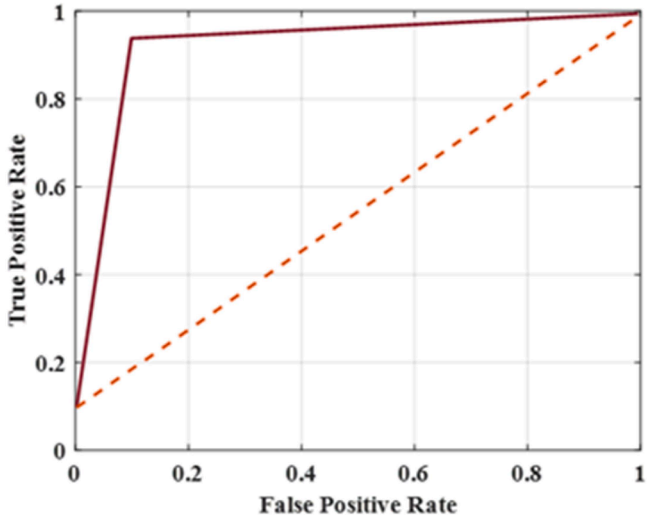


Fig. 10. ROC curve.

CRediT authorship contribution statement

K. Praveen Kumar: Writing – review & editing. **Dr. N. Suresh Kumar:** Writing – original draft, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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